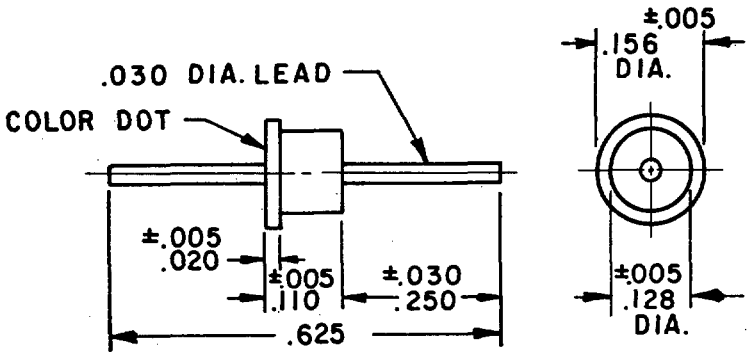
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE),
(FEED-THRU)
STYLE CK99

This specification is approved for use by Aeronautical Systems Division, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-11015.






CIRCUIT DIAGRAM



Approximate size

| INCHES | MM | INCHES | MM |
|---|---|---|---|
| .005 | .13 | .128 | 3.25 |
| .020 | .51 | .156 | 3.96 |
| .030 | .76 | .250 | 6.35 |
| .110 | 2.79 | .625 | 15.88 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.015 (.38 mm).

FIGURE 1. Style CK99 capacitors.



FSC 5910

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:59Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1.
Case type - Feed-thru (solder mount).
Capacitance value - See table I.
Capacitance tolerance - ±20% (M).
Rated temperature - -55° to +125°C.
Rated voltage - See table I.

Dielectric withstanding voltage:
Dielectric (DWV): Method 301 of MIL-STD-202.
Test voltage -200 percent of dc rated voltage applied between the terminal and the mounting flange.
Body insulation: Not applicable.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition C (70,000 ft).
Test potential - 200 percent of dc rated voltage applied between the terminal and the mounting flange.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 500 megohms, min, measured between the terminal and the mounting flange.

Capacitance (Cap.): Method 305 of MIL-STD-202.
Test frequency - 1.0 ±0.1 kHz.
Test voltage - 1.0 ±0.2 volt rms.

Dissipation factor (DF): 5.0%, max,

Shock (specified pulse): Method 213 of MIL-STD-202, condition A (50 G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G).
Mounting - Capacitors shall be rigidly mounted by the body.

Thermal shock and immersion:
Thermal shock: Method 107 of MIL-STD-202, condition A.
Immersion: Method 104 of MIL-STD-202, condition A.
IR - 50 megohms, min.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition A.

Terminal strength: Not applicable.

Moisture resistance: Method 106 of MIL-STD-202.
IR - 50 megohms, min.
ΔCap. - Within ±10% of initial measurement.

Solderability: Method 208 of MIL-STD-202 (both ends).

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.

Voltage-temperature limits: In accordance with MIL-C-11015 (symbol BW).

Life (at elevated ambient temperature):
Method 108 of MIL-STD-202.
2,000 hours (for qualification): 250 hours (performance check): 1,750 hours continuation test).
Test potential - 200% of rated voltage.
IR - 100 megohms, min (at 85°C);
250 megohms, min (at 25°C).
DF - 8.0% max (at 25°C).

Marking: Capacitors shall be marked as shown on figure 1. See table 1 for color code.
The type designation, capacitance, voltage, and the manufacturer's code or symbol shall be marked on the package container.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:59Z
Check the source to verify that this is the current version before use.

TABLE I. Style CK99 characteristics.

| Type designation | Capacitance value | Rated voltage | Color dot |
|---|---|---|---|
| | (pF) | (volts, dc) | |
| CK99BW101M | 100 | 200 | Green |
| CK99BW501M | 500 | 200 | Brown |
| CK99BW122M | 1,200 | 200 | White |
| CK99BW272M | 2,700 | 100 | Red |
| CK99BW502M | 5,000 | 50 | Yellow |

Preparing activity:
Air Force - 11

Agent:
DLA - ES

(Project 5910-F322)

☆U.S. GOVERNMENT PRINTING OFFICE: 1977 — 703-122/6



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:59Z
Check the source to verify that this is the current version before use.